# Anisha Saha

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## Research Interests

I work at the intersection of **multimodal understanding and reasoning**, studying how language, audio, and visual signals interact to convey intent. My research focuses on multimodal information integration in LVLMs, open-ended reasoning and evaluation of **robustness, bias, and hallucination in LLMs and LVLMs**.

## Academic & Professional Background

2024 – Present — **Max Planck Institute for Informatics**, SIC, Germany.
*ELLIS PhD Student, Multimodal Language Processing*
Advisors: Prof. Dr. Vera Demberg & Prof. Dr. Timothy Hospedales

2022 – 2024 — **Micron Technology, Inc.**, Hyderabad, India.
*Data Scientist*

2020 – 2022 — **Chennai Mathematical Institute**, India. CGPA : 9.44
*M.Sc. Data Science*

2017 – 2020 — **St. Xavier's College**, Kolkata, India. CGPA : 7.27
*B.Sc. Mathematics*. Dissertaion: Integrating Factors of 1st order ODEs.

## Publications

- **MUStReason: A Benchmark for Diagnosing Pragmatic Reasoning in Video-LMs for Multimodal Sarcasm Detection** [Link]
  Anisha Saha, Varsha Suresh, Timothy Hospedales, Vera Demberg
  *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*
- **System-Mediated Attention Imbalances Make Vision-Language Models Say Yes** [Link]
  Tsan Tsai Chan, Varsha Suresh, Anisha Saha, Michael Hahn, Vera Demberg
  *ACL 2026 (Findings)*
- **M$^3$QuestionIng: Multi-modal Multi-span Medical Question Answering**
  Anisha Saha, Vaibhav Rathore, Abhisek Tiwari, Akash Ghosh, Sai Ruthvik Edara, Sriparna Saha
  *ACM Transactions on Computing for Healthcare (HEALTH)*
- **Experience and Evidence are the eyes of an excellent summarizer! Towards Knowledge Infused Multi-modal Clinical Conversation Summarization** [Link]
  Abhisek Tiwari, Anisha Saha, Sriparna Saha, Pushpak Bhattacharya, Minakshi Dhar
  *ACM Conference on Information and Knowledge Management (CIKM 2023)*

## Research Experience

- **Indian Institute of Technology, Patna.** Mentor: Prof. Dr. Sriparna Saha
  *Built multimodal multi-tasking frameworks for clinical dialogue summarization, medical domain identification, dialogue generation and medical question-answering.*
- **Indian Institute of Information Technology, Hypderabad.**
  Mentor: Prof. Dr. Ponnurangam Kumaraguru
  *Model augmentation for improving downstream performance on code-switched tasks.*
- **Chennai Mathematical Institute.** Mentor: Prof. Dr. Venkatesh Vinayakarao.
  *Explored distributed representations for code-to-vector embeddings and Hierarchical Attention Networks.*

## Internships

- **Data Science Intern,** Delab Research Pvt. Ltd.
  *Deployed deep-learning architectures for a production-level smart energy monitoring system and day-ahead solar energy forecasting.*
- **Julia Statistics Developer,** XKDR Forum.
  *Contributed to the Surjey.jl package in Julia, required to study complex survey data.*
- **Summer Intern (Digital & AI),** Carelon Global Solutions.
  *Generated statistical quality control parameters for models processing audio transcripts, performed root cause analysis to address data anomalies and evaluate model performance.*

## Teaching Experience

- **Teaching Assistant, Statistics Lab**, Saarland University .
- **Teaching Assistant, Linear Algebra and Applications**, Chennai Mathematical Institute .
- **Teaching Assistant, Computing and Data Science**, Sai University .

## Projects

- **Diabetic Retinopathy Detection**
  Created an automated analysis system capable of classifying each retinal image to one of the 5 classes, corresponding to the severity of the disease.
- **Day Ahead Solar Photovoltaic Forecasting of Power Output : A Comparative Study**
  Performed a comparative evaluation of the prediction models, generated using the techniques of ARIMA, SVM, Deep Nets and XGBoost, for forecasting day ahead power output of Photovoltaic systems. Evaluating ML models against traditional time series forecasting techniques. Identification of relevant features for day ahead energy production.
- **Facial Recognition using Eigenfaces**
  Compared characteristics of a face image to that of face images of known individuals and non-face images. Treated the images as two-dimensional, rather than requiring three-dimensional geometry, by taking advantage of the fact that faces are generally upright. Eigenfaces method was used to project the images into a low-dimensional feature space using Principal Component Analysis.
- **Unreeling Netflix**
  Built a movie recommendation engine on Apache PySpark. Demonstrated how Netflix uses a recommendation system based on Big Data to recommend movies to its users. Focused on collaborative vs content-based filtering for recommendation.
- **Integrating Factors of 1st order Ordinary Differential Equations**
  This was my Bachelor's dissertation. Depicted the correlation between the integrating factors of a first order ordinary differential equation and infinitesimal Lie group transformations. Corresponding to each integrating factor of an Ordinary Differential Equation (ODE), a symmetry group can be found, under which the ODE is form-invariant.